Impaired neuromusculoskeletal response to training stimuli associated with low energy availability: a systematic review

Alexandra F DeJong Lempke ,[1,2] Katherine L Smulligan,[3,4] Gauri A Desai,[5] Kelsey E Hagan,[2,6] Jessie R Oldham ,[1,2] Leila Z Islam,[6] Kristin E Whitney [3,4]

► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/bjsports-2025-110096).

[1]Physical Medicine and Rehabilitation, Virginia Commonwealth University School of Medicine, Richmond, Virginia, USA
[2]Institute for Women's Health, Virginia Commonwealth University, Richmond, Virginia, USA
[3]Boston Children's Hospital, Boston, Massachusetts, USA
[4]Harvard Medical School, Boston, Massachusetts, USA
[5]Orthopedic Surgery, Stanford University, Stanford, California, USA
[6]Psychiatry, Virginia Commonwealth University School of Medicine, Richmond, Virginia, USA

Correspondence to
Dr Alexandra F DeJong Lempke; Alexandra.lempke@vcuhealth.org

Accepted 25 September 2025

Check for updates

© Author(s) (or their employer(s)) 2025. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: DeJong Lempke AF, Smulligan KL, Desai GA, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2025-110096

ABSTRACT

Objective Low energy availability (LEA) impairs musculoskeletal health, with emerging evidence of impaired neuromusculoskeletal adaptations to training. We aimed to synthesise the existing evidence examining neuromusculoskeletal responses to training stimuli among individuals with LEA.

Design Systematic review (PROSPERO 603258).

Data sources Six databases were searched on 4 November 2024 (total n=6399; duplicates n=2811; articles to screen n=3388). Search terms included the population and training-related exposures. Two blinded, independent reviewers screened articles using the Covidence web-based tool.

Eligibility criteria Original articles written in English with any form of neuromusculoskeletal responses to training stimuli (premeasures and postmeasures) assessed among individuals aged ≥10 years with LEA were included. Study details, methodological quality (Physiotherapy Evidence Database, PEDro scale) and descriptive summaries with change percentages by time point and LEA status were extracted.

Results 21 studies met inclusion criteria (total participants: 536; 44% males, 56% females; PEDro: 5±2). Inconsistencies in LEA thresholds and criteria, exercise exposures and follow-up time frames precluded pooled analyses. All studies examined changes in lean muscle or fat-free mass (FFM), with seven also examining functional or activity-specific performance, and fewer concurrently examining strength, limb circumference, cellular-level or subjective neuromusculoskeletal measures. 10 studies identified that those with LEA had impaired lean mass/FFM responses (−1% to −5%), and 10 found no substantial changes over time (<1%). Six studies identified functional/sport-specific impairments over time among those with LEA (−4% to −10%). All remaining primary adaptations (strength, limb circumference, cellular-level and subjective measures) were impaired with LEA. These findings are clinically meaningful to consider, particularly for LEA-related impaired responses to rehabilitation targeting muscle hypertrophy and strength.

Conclusions Coaches, athletes and clinicians should be aware of the evidence suggesting LEA impairs neuromusculoskeletal training responses.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Low energy availability (LEA) has been linked with musculoskeletal health concerns, such as increased injury risk and worse muscle strength compared with those with optimal energy availability. Emerging evidence suggests that LEA may additionally impair the neuromusculoskeletal response to training stimuli and exacerbate health concerns, yet there is no current evidence synthesis on these potential ramifications.

WHAT THIS STUDY ADDS

⇒ Current evidence supports that those with LEA have either no significant changes or decreased lean muscle or fat-free mass in response to training over time, suggesting that athletes with LEA may not have desired muscle hypertrophy effects and may, in fact, exhibit muscle loss in response to training. Individual studies identified functional and/or sport-specific neuromusculoskeletal impairments to training stimuli among those with LEA. Specifically, individuals with LEA presented with reduced muscle protein synthesis, reduced limb circumference and reduced muscle strength in response to training.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This systematic review supports the need for future comprehensive neuromusculoskeletal response assessments among those with LEA to design targeted interventions to preserve health and improve intended responses to training stimuli.

INTRODUCTION

Low energy availability (LEA) refers to a discrepancy between energy intake and energy expenditure from physical activity, resulting in inadequate energy stores to support basic physiological functioning.[1–6]

LEA is a prevalent concern among physically active individuals, as at least one in four physically active females and one in six physically active males do not achieve adequate fueling and are in a state of LEA.[7–13] Endurance athletes are at particularly high risk of developing LEA, with prevalence as high as 75% noted in past research.[8–11 14] There are a host of potential factors contributing to underfuelling and LEA, such as eating disorders/disordered eating behaviours, inappropriate or intentionally restricted macronutrient and micronutrient consumption, and inadvertent overexercising without adequate calorie replacement (ie, underfuelling). Problematic LEA, which is a prolonged and/or severe mismatch

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

between energy intake and exercise energy expenditure, may be associated with the manifestation of Relative Energy Deficiency in Sport (REDs),[4 15 16] involving a constellation of impairments in physical functioning across multiple body systems, particularly the central nervous,[17–20] endocrine,[1 21] genitourinary/reproductive systems[22–28] and neuromusculoskeletal systems.[29–36] Disruptions to body systems stemming from LEA synergistically lead to impaired physical health and overall athletic performance. Emerging evidence has illuminated the role of LEA in impaired neuromusculoskeletal health and response to imposed physical training stimuli.[1 4 28 33 36]

Recent literature suggests a musculoskeletal resistance to imposed training demands potentially stemming from LEA and contributing to orthopaedic health concerns among physically active individuals. It is well-established that individuals with LEA are at heightened risk of bone stress injuries (BSI),[33 34] with a four-fold higher risk among females with LEA versus those without.[33] This heightened BSI rate is a postulated consequence of bone tissue weakening due to inadequate macronutrient and micronutrient consumption evidenced by low bone mineral density[37 38] and poor shock absorption capabilities of surrounding musculature.[39–43] Accordingly, the number of studies examining neuromusculoskeletal system responses among individuals with LEA has increased. Preliminary evidence suggests that temporarily inducing LEA during resistance training impedes cellular-level (muscle protein synthesis) to functional-level (strength, performance) muscle responses, spanning from short (10 days) to long-term time frames (20 weeks—greater than 1 year).[4 29–31 44] Individuals with LEA are already expected to be at a neuromusculoskeletal deficit compared with those with adequate energy availability, and this emerging evidence suggests that this includes inadequate neuromusculoskeletal responses to imposed training stimuli. However, there is currently no synthesised information examining neuromusculoskeletal responses to training among individuals with LEA. Collating this information will help athletes, especially those at risk for LEA development, and key athlete allies to better understand the risks associated with underfuelling, and ultimately develop programming geared towards improving energy balance to support athlete health and performance. Understanding neuromusculoskeletal responses is especially important during clinical efforts aimed at increasing muscle mass where LEA may stand to impair patient recovery, such as among patients completing neuromusculoskeletal rehabilitation following major injury or surgery (ie, anterior cruciate ligament reconstruction).

The purpose of the present systematic review was to synthesise the existing evidence examining neuromusculoskeletal responses to imposed training stimuli among individuals with LEA. Training stimuli were included across a broad range of categories, from controlled strengthening or endurance training programmes to sporting programmes, to capture the breadth of available literature on this topic area for synthesis. Neuromusculoskeletal responses of interest ranged from tissue-level to functional performance responses. We hypothesised that most published studies would reflect impaired neuromusculoskeletal responses to training stimuli over time, and/or dampened neuromusculoskeletal responses under LEA conditions compared with conditions without LEA.

MATERIALS AND METHODS

The research team conducted a systematic review in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (figure 1).[45] The protocol was submitted to the international prospective register of systematic reviews (PROSPERO, 603258) on 24 October 2024 and was officially registered on 4 November 2024. The research team consulted with a university librarian to conduct a robust search in alignment with the systematic review objective and primary research question: 'How does the presence of LEA influence neuromusculoskeletal response to training stimuli?' The search terms, strategy and approach to data extraction and presentation were mutually agreed on by the research team prior to initiating the project. A web-based systematic review tool (Covidence, Veritas Health Innovations, Melbourne, VIC, Australia) was used to manage and conduct all systematic review and study selection elements.

Search strategy

The research team and librarian consulted developed the search strategy using a Population, Intervention, Comparison(s), Outcome(s) (PICO) framework. The target population was individuals who met specific study definitions of LEA, and interventions were inclusive of any strengthening, aerobic, physical conditioning or sports participant/sport-specific training programmes without restriction on the level of athletic competition or ability. Exact definitions and thresholds for LEA status from each study were extracted and summarised in table 1. LEA was intentionally broadly defined for the purpose of this systematic review given that there is no one universally accepted threshold of LEA through direct measurements (kcal/kg/day), and as LEA indicators through validated health questionnaires are sometimes used to define LEA status.[4 46 47] Comparisons were primarily neuromusculoskeletal adaptations in response to training over time compared with baseline within those with LEA, though control participant comparisons without LEA were also considered. The outcomes of interest were any neuromusculoskeletal adaptations, for example, but not limited to, fat-free mass (FFM) or muscle strength/power. The only elements of the PICO format used in the official search were the Population and Intervention, as the research team did not want to pre-emptively limit neuromusculoskeletal outcomes of interest, which were used to guide the article screening phase of the systematic review. The full search strategy is presented in figure 2. The terms were searched in the PubMed Medical Literature Analysis and Retrieval System Online, Excerpta Medica Database (Embase), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Web of Science, SPORTDiscus and Cochrane databases on 04 November 2024.

Potential articles were included if they were written in the English language, examined any neuromusculoskeletal outcomes following a physical training intervention in an LEA population, and included adolescent and/or adult participants (age ≥10 years). The specific interventions of interest for this review were any form and duration of physical strengthening, conditioning or exercise intervention/therapy. The research team did not pre-emptively limit the timeframe, duration, volume or intensity of the intervention to get an accurate representation of the available literature. Interventions were included if they were delivered in person, or remotely (asynchronously or synchronously). Potential studies were excluded if they were conducted in the wrong population (other physiological conditions aside from LEA or synonymous condition terms, <10 years), did not specifically examine the effects of a physical intervention, or did not report at least one pre–post or between-group (LEA vs non-LEA) measure of neuromusculoskeletal outcomes. Other systematic reviews and meta-analyses, literature reviews, scoping reviews,

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

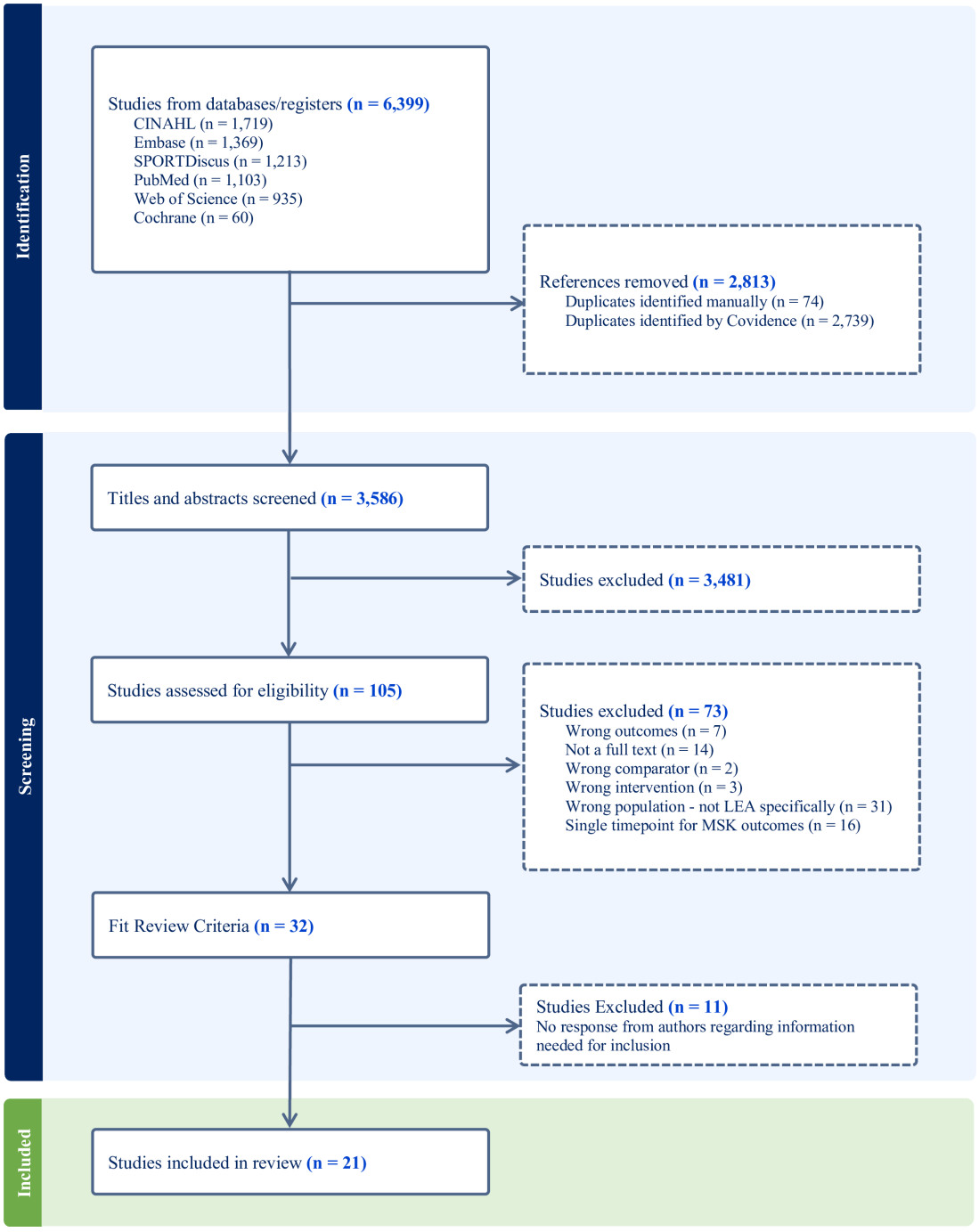


Figure 1 PRISMA study selection and screening flow chart. Study identification process by stage and rationale for exclusion. All studies were reviewed by two independent reviewers to determine eligibility. This flow chart was auto-generated using the Covidence web-based systematic review tool (Covidence, Veritas Health Innovations, Melbourne, VIC, Australia). PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

case studies and case series, conference abstracts, animal studies and cadaveric studies were excluded.

Study selection

Each article identified in the initial search was independently screened by two of the five authors, with all authors serving as independent, blinded reviewers in pairs (AFDL, KEH, JRO, LZI and GAD). The reviewers screened studies by title, then abstract and finally the full text to determine eligibility for inclusion in this systematic review. Disagreements between the reviewers were first discussed interpersonally to determine if a consensus could be reached, and, as needed, a senior independent reviewer made the tie-breaking decision (KEW). The study selection decision and reason for exclusions were noted within Covidence (figure 1).

Data extraction

The following data were extracted from each included article: first author, year, journal, country, setting, study design, inclusion and exclusion criteria, LEA determination, total participants enrolled, total participants analysed, attrition, participant sex, participant age, physical training characteristics, time between assessments, primary outcomes related to neuromusculoskeletal adaptations

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 1 Study, design, participant details and low energy availability determination details of the included studies

Author and year; Journal	Country and setting	Study design	Low energy availability determination
Alemany 2008; J Appl Physiol[60]	USA; Marine Corps Infantry Officer Candidate School	Randomised crossover trial	Directly manipulated by investigators. Both groups had low energy intake (6.5 MJ/day). Random assignment to either a high protein low energy diet or low protein low energy diet (Diet type not included for this review)
Baranauskas 2017; J Am Coll Nutr[71]	USA; University Athletics	Prospective cohort study	EA=(EI–EEE(over 8 weeks))/FFM All athletes had LEA defined as ≤30 kcal/kg FFM/day
Barreto 2021; Int J Sports Physiol Perform[73]	Brazil; Professional Cycling	Prospective cohort study	EA=(EI–EEE(over 2 days))/FFM. LEA: <30 kcal/kg FFM/day Suboptimal EA: 30–45 kcal/kg FFM/day Optimal EA: >45 kcal/kg FFM/day
Burke 2023; Med Sci Sport Exerc[72]	Australia; Race Walker Training Camp	Non-randomised controlled trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA group: 15 kcal/kg FFM/day High carbohydrate/high energy group: 45 kcal/kg FFM/day
Friedl 1994; J Appl Physiol[61]	USA; Army Research Institute of Environmental Medicine	Prospective cohort study	Calculated from meal rations and energy expenditure using double-labelled water. Estimated energy deficit of 5.0±2.0 MJ/day (1.4–10.7 MJ/day).
Gifford 2021; Med Sci Sport Exerc[62]	UK; British Army Officer Commissioning Course at the Royal Military Academy	Prospective cohort study	EA=(EI–EEE(over 10 days))/FFM. Completed for total physical activity, and moderate-vigorous physical activity. LEA: <30 kcal/kg FFM/day (reference to other literature only) Secondary: BEDA-Q responses
Ishibashi 2020; Physiol Rep[63]	Japan; Sports Science Institute	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA condition (4 days): <20 kcal/kg FFM/day Control condition (4 days): >45 kcal/kg FFM/day
Jeppesen 2024; Redox Biol[64]	Denmark; University	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA condition (14 days): 22 kcal/kg FFM/day Control condition (14 days): 50 kcal/kg FFM/day
Jesus 2022; Int J Sport Nutr Exerc Metab*[14]	Portugal; Sports federations, clubs and high-performance centres near University	Cohort study	EA=(EI–EEE(over 7 days))/FFM LEA: <30 kcal/kg FFM/day Subclinical LEA: (M) 30–40 kcal/kg FFM/day, (F) 30–45 kcal/kg FFM/day Optimal EA: (M) >40 kcal/kg FFM/day, (F)>45 kcal/kg FFM/day Note: Only triathletes had a critical number of LEA for this review
Jouhki 2024; Scand J Med Sci Sports[75]	Finland; University	Randomised controlled trial	EA=(EI–EEE(3 days))/FFM. Acknowledged <30kcal/kg FFM/day cut-off value, but did not use a universal cut-off for the study. NOTE: use of diet group only who met this threshold between timepoints
Jurov 2022; J Int Soc Sports Nutr[76]	Slovenia; National Cycling/ Triathlon, Professional Cycling	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE(7 days))/FFM. Step 1 LEA (14 days): 25% reduction to 22.4±6.3 kcal/FFM/day Step 2 LEA (14 days): 50% reduction to 17.3±6.3 kcal/FFM/day Step 3 LEA (10 days): 75% reduction to 8.8±3.3 kcal/FFM/day
Jurov 2022; Eur J Appl Physiol[74]			Directly manipulated by investigators. EA=(EI–EEE(7 days))/FFM. LEA condition (14 days): 25% reduction to 22.4±6.3 kcal/FFM/day
Kojima 2022; PLOS ONE[65]	Japan; University	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA condition (2 days): 15 kcal/kg FFM/day Control condition (2 days): 45 kcal/kg FFM/day
Kojima 2020; Med Sci Sport Exerc[66]	Japan; University	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA condition (3 days): 15 kcal/kg FFM/day Control condition (3 days): 45 kcal/kg FFM/day
Kuikman 2025; Med Sci Sport Exerc[67]	Australia; University	Randomised crossover trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. 5 conditions of a combination of energy availability with/without exercise. For this review: LEA condition (24 hours): 15 kcal/kg FFM/day High EA condition (24 hours): 45 kcal/kg FFM/day
Mathisen 2020; Scand J Med Sci Sports*[68]	Norway; Federation of Bodybuilding and Fitness	Cohort study	LEAF-Q for measuring severity of LEA status (≥2 for gastrointestinal dysfunction, 4 for menstrual irregularities, 8 for total LEAF-Q). All female athletes were categorised as LEA per author contact.
Oxfeldt 2023; J Physiol[29]	Denmark; University	Randomised controlled trial	Directly manipulated by investigators. EA=(EI–EEE)/FFM. LEA group (10 days): 25 kcal/kg FFM/day Optimal EA group (10 days): 50 kcal/kg FFM/day
Oxfeldt 2024; Med Sci Sport Exerc[69]			
Schaal 2017; Scand J Med Sci Sports[70]	France; National Synchronised Swim Team	Cohort study	EA=(EI–EEE(over 4 days))/FFMLEA definition: ≤30 kcal/kg FFM/day
Tektunalı Akman 2024; Front Public Health*[77]	Turkey; Adolescent Sport Teams	Randomised controlled trial	EA=(EI–EEE(over 3 days))/FFMLEA definition: ≤30 kcal/kg FFM/day Reduced EA definition: 30–45 kcal/kg FFM/day Optimal EA definition: ≥45 kcal/kg FFM/day
Zabriskie 2019; Nutrients*[78]	United States; University Athletics (Lacrosse)	Cohort study	EA=(EI–EEE(over 4 days))/FFMLEA definition: ≤30 kcal/kg FFM/day

All standard journal abbreviations included and thus not individually defined.
*Indicates that authors were contacted for included data.
†Extracted data from the included studies pertaining to key study and participant details.
BEDA-Q, Brief Eating Disorder in Athletes Questionnaire; EA, Energy Availability; EDE-Q, Eating Disorder Examination Questionnaire; EEE, Estimated Energy Expenditure; EI, Energy Intake; Exc, Exclusion; F, Females; FFM, Fat-Free Mass; Inc, Inclusion; LEA, Low Energy Availability; LEAF-Q, Low Energy Availability in Females Questionnaire; M, Males; NCAA, National Collegiate Athletic Association.

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

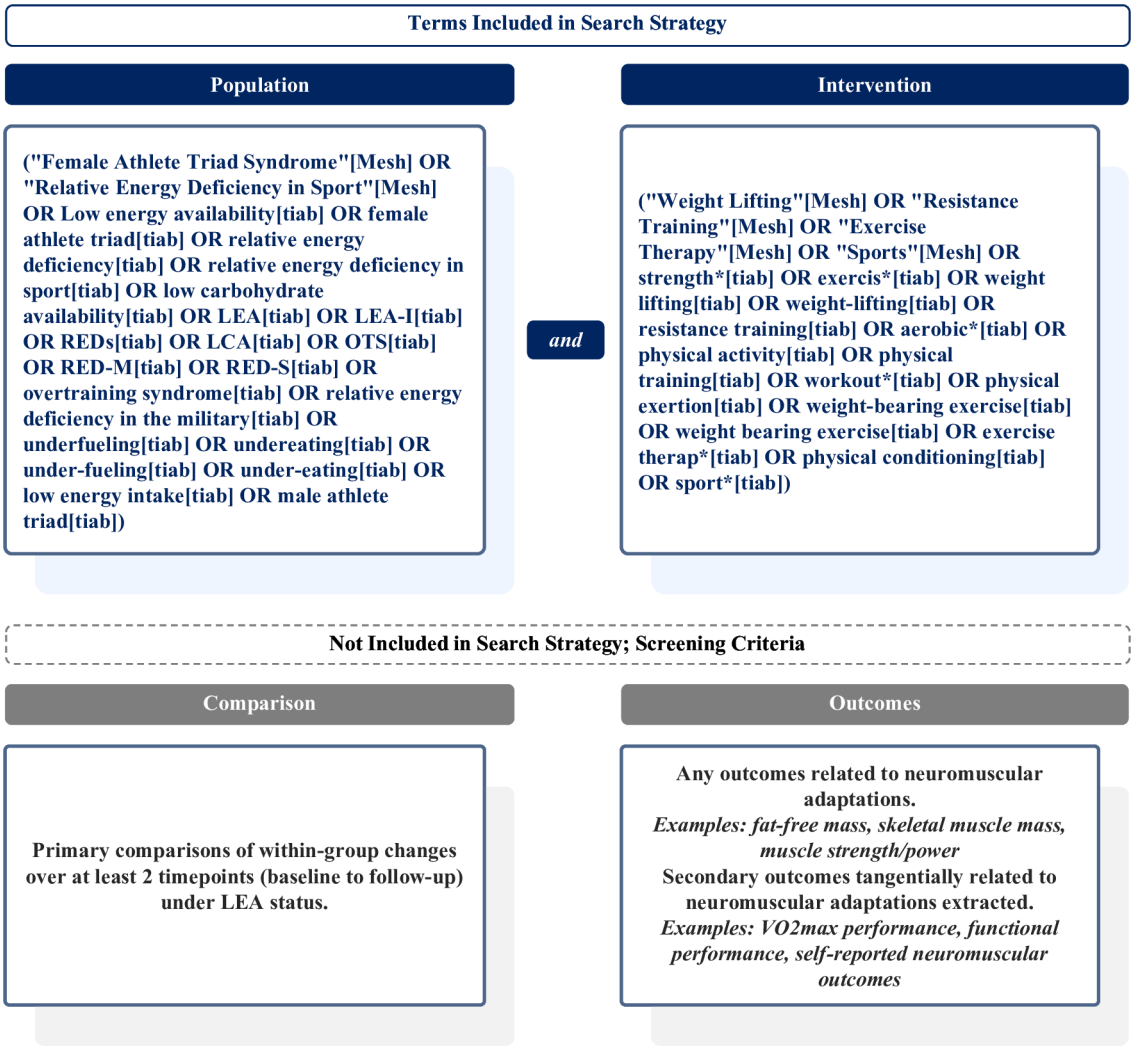


Figure 2 Search terms and screening criteria. Search terms used to identify potential articles across search engines. Presented search terms are those used for the PubMed search, but a similar approach with engine-specific subject headings, searches through titles, abstracts and other fields, and alternative endings was used across platforms. *Signifies that word alternative endings were searched for the specific term.

and associated findings, and secondary outcomes related to neuromusculoskeletal adaptations and associated findings. One reviewer extracted the data per study, though two reviewers were assigned to complete methodological quality assessment for each article (see below). The lead author (AFDL) contacted the authors of the included studies in the event of data missingness or the need for additional information for the included studies. Studies that did not explicitly define inclusion or exclusion criteria were still maintained in this review as long as LEA determination and key outcomes pertaining to neuromusculoskeletal measures were present or able to be obtained from contacted authors.

Methodological quality assessment

Each included study was assessed for methodological quality by two independent, blinded reviewers using the Physiotherapy Evidence Database (PEDro) scale.[48 49] Quality score disagreements were handled in the same manner as described at the study selection stage.

Determination and plan for pooled analyses

We did not anticipate the ability to synthesise data given heterogeneity in training approaches/possible outcomes. The following criteria were used to determine if data synthesis was possible: The same definition of LEA study inclusion criteria, at least four studies that employed the same mode, number, dosing and type of training intervention, and the same outcome measure target structure, type and measurement units. If pooled analyses were possible, the change scores in outcome measures specific to neuromuscular adaptations (by type/muscle group) would be assessed using a random effects meta-analysis using R software.

Statistical analyses

Formal statistical analyses of pooled assessments were not possible due to the criteria specified above. Instead, a narrative synthesis of the included studies was conducted for the purpose of this systematic review. Only descriptive summaries were used to classify the included study characteristics. When possible, the percentage of changes across time points (baseline to follow-up(s)) and/or conditions (LEA vs non-LEA designations) was calculated from the mean data from the articles. For studies that incorporated interventions other than training interventions (ie, specialised diets, nutrition education), only the control group change data were extracted for this review to remove the potential influence of these controlled factors on the neuromusculoskeletal outcomes versus LEA status alone.

EQUITY, DIVERSITY AND INCLUSION STATEMENT

The authors included on this review are all females, represented across varying career stages (postdoctoral fellows, assistant

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

professors, physicians) and across three different institutions in the USA. Studies included in this review were from 13 different countries and included both female and male cohorts (tables 1 and 2). Most participants were young adults, and the results should be interpreted accordingly.

RESULTS

The initial search on 4 November 2024 yielded 6399 total results, with 2813 duplicates across search platforms; thus, there were 3586 articles to screen (figure 1). There were 3481 studies deemed irrelevant at the title and abstract screening phase, and 73 full-text articles removed (reasons for exclusion provided in figure 1). Authors of 15 studies were contacted to obtain relevant data (ie, neuromusculoskeletal outcomes over time by LEA status). However, 11 did not respond, and as such, the studies were excluded.[12 50–59] Therefore, 21 studies were included in this review (total participants: 536; 44% males, 56% females; table 1).[14 29 60–78] Agreement among reviewers was an average of 92%, with a consensus reached during discussions among reviewers for 99%, leaving 1% for a final decision from the senior author. The methodological quality of the included studies was overall low, with no studies achieving the maximum score of 10 (mean PEDro scores: 5±2; minimum: 2, maximum: 9; table 2).

The majority of studies (13/21) defined LEA through direct assessments or manipulations of energy availability through the formula:[79]

$$\textit{Energy Availability}\ [kcal/kg/day] = \frac{(\textit{Energy Intake}\ [kcal] - \textit{Estimated Energy Expenditure}\ [kcal])}{\textit{Fat Free Mass}\ [kg]}$$

However, the included studies used varying LEA thresholds: <15 kcal/kg/day,[65–67 72] <20–25 kcal/kg/day[29 63 64 69] and <30 kcal/kg/day.[14 62 70 71 73 75 77 78] Additional methods of defining LEA included millijoules of daily energy intake,[60 61] percentage reductions of baseline energy availability[74 76] and one study used questionnaire-based criteria to assess for indicators of LEA (table 1).[68]

Nine studies directly controlled the exercise training exposure[29 63 64 66 67 69 74 76] and were most consistently endurance-based training programmes (cycling, running) at a percentage of maximal aerobic capacity measured at baseline (online supplemental table 1). Three studies were integrated with standardised military training courses,[60–62] while the remaining nine studies recorded exercise exposures through sport or activity involvement inherent to sport-specific training of the target cohorts (online supplemental table 1).[14 68 70–73 75 77 78] Length of training exposure and length of follow-up varied from immediate effects (12 hours to several days),[63 65–67] moderate-term effects (1–4 weeks),[29 60 64 69 70 72 74 76] to longer-term effects (multiple months to years) between assessments.[14 61 62 68 71 73 75 77 78]

Of the included studies, all examined lean or FFM or skeletal muscle mass,[14 29 60–78] seven examined activity-specific or functional performance,[62 64 69 72–74 76] and four examined cellular-level neuromusculoskeletal outcomes (glycogen, muscle protein synthesis) (online supplemental tables 1 and 2).[29 63 66 69] Few studies examined muscle strength,[69] limb circumference for muscle girth[61] and self-reported muscle outcomes (ie, fatigue, soreness).[66 70] Secondary, tangentially related neuromusculoskeletal response outcomes included maximal aerobic capacity (VO_2max)[65 66 72–74 76] and blood glucose (online supplemental table 1).[65 66 69]

Lean or FFM responses to training

Full body lean mass and FFM were assessed using standard dual X-ray absorptiometry (DXA) methods for body composition analysis. 10 studies identified that those with LEA classifications had decreased FFM or lean muscle mass over time compared with baseline measures (online supplemental tables 1 and 2).[29 60–64 66 67 69 76] These decreases ranged in magnitude from 1% to 5% change over time. Only one study identified a significant increase in FFM over time,[68] while all remaining articles found no significant change in FFM/lean muscle mass associated with LEA over time (within 1%; online supplemental tables 1 and 2).[14 65 70–75 77 78] There were no consistent methodological factors that influenced null versus significant lean or FFM findings (ie, duration of training, length of follow-up, LEA thresholds, study design).

Functional or activity-specific responses to training

Functional activities included countermovement jump (CMJ) performance,[69 74 76] agility tests[69 73 74 76] and sport-specific time or distance trials (online supplemental table 1).[62 64 69 72 73] Five of the seven studies identified that functional or activity-specific responses were significantly impaired over time within LEA classifications (online supplemental tables 1 and 2).[62 64 69 73 74 76] The only outcome not found to be significantly impaired was race walking performance over time (online supplemental tables 1 and 2).[72]

Additional neuromusculoskeletal outcomes

Although the remaining results were identified among few or singular studies, all remaining results indicated that neuromusculoskeletal responses were impaired over time among those with LEA classifications (online supplemental tables 1 and 2). Specifically, LEA was found to impair expected muscular strength gains[69] and regional circumference (girth) measures over time,[61] was associated with more fatigue[70] and muscle soreness over time,[66] and led to reduced muscle protein synthesis[29] over time (online supplemental tables 1 and 2).

For secondary outcomes tangentially related to neuromusculoskeletal adaptations, all studies that examined maximal aerobic capacity either through cycling or running tests identified impaired responses, particularly with greater rises in lactate concentration, over time among individuals with LEA.[65 66 72–74 76] Furthermore, blood glucose[29 65 66] and muscle glycogen content[63 66 69] were found to be impaired over time among those with LEA classifications (online supplemental tables 1 and 2).

DISCUSSION

There is emerging evidence exploring the effects of exposure to LEA on neuromusculoskeletal health, and subsequent responses to physical training and activity over time.[14 29 60–78] This review aimed to systematically synthesise the current evidence to determine if neuromusculoskeletal responses to training stimuli were impaired and potential threats to physical performance associated with LEA. Overall, most of the current evidence suggests that functional or sport-specific adaptations are significantly impaired across a range of assessment time frames among those with LEA. Lean mass or FFM had the most conflicting findings, with approximately half of the included studies reporting specific impairments associated with LEA, and approximately half failing to identify impairments over time. Preliminary evidence also suggests that cellular-level and functional-level muscle responses may be impaired under LEA conditions. However,

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 2 Methodological quality and participant details for the included studies

Author and year; Journal	PEDro score	Inclusion/exclusion Criteria	N enrolled/ N Analysed (Attrition %)	Sexes (male/female)	Ages (years)
Alemany 2008; J Appl Physiol[60]	7	Inc: Not defined. Exc: Not defined	36/34 (94%)	34 M/0 F	24.5±0.3
Baranauskas 2017; J Am Coll Nutr[71]	4	Inc: Indoor track athletes competing in events ranging from 800 to 5000. Exc: Known metabolic condition.	24/12 (50%)	8 M/4 F	20.5±1.4
Barreto 2021; Int J Sports Physiol Perform[73]	2	Inc: Top-level Brazilian female cyclists from the same professional cycling team Exc: Not defined	5/4 (80%)	0 M/4 F	26±4
Burke 2023; Med Sci Sport Exerc[72]	5	Inc: Highly trained race walkers completing a research-embedded training camp. Exc: Not defined	23/22 (96%)	19 M/3 F (LEA: 10 M/1 F)	LEA: 29.3±4.5
Friedl 1994; J Appl Physiol[61]	3	Inc: Male volunteers from a single class of an Army Ranger Training Course Exc: Not defined	190/55 (29%)	55 M/0 F	23.6±2.8
Gifford 2021; Med Sci Sport Exerc[62]	4	Inc: Women in the British Army Officer Commissioning Course who met exacting medical standards mandated before employment in the Army Exc: Diagnosed thyrotoxicosis, eating disorder, malabsorption or food intolerance.	68/47 (69%)	0 M/47 F	23.9±2.6
Ishibashi 2020; Physiol Rep[63]	7	Inc: Male long-distance runners completing regular training (6 days/week, 4 hours/day). Exc: Smoking, use of herbal medicine, supplements, medications	6/6 (100%)	6 M/0 F	19.8±0.4
Jeppesen 2024; Redox Biol[64]	9	Inc: Females aged 18–40 years, BMI 18–23 kg/m^2, regular normal menstrual cycle or hormonal contraceptive users, maintained body mass (±3 kg) and >6 hours of endurance exercise every week for 6 months preparticipation. Exc: Ratio between measured and predicted RMR of <0.90, smoking, chronic use of prescription medication, chronic illness or history of eating disorders.	12/12 (100%)	0 M/12 F	26.8±3.4
Jesus 2022; Int J Sport Nutr Exerc Metab*[14]	4	Inc: Athletes training >10 hours/week, negative performance enhancing drug tests, not taking medications/dietary supplements Exc: Use of oral contraceptives or non-eumenorrhoea for females	88/56 (64%)	39 M/17 F	19.1±4.2
Jouhki 2024; Scand J Med Sci Sports [75]	7	Inc: Healthy adults aged 20–40 years and with at least 2 years of resistance training experience, registered under the national doping control and testing organisation Exc: Chronic diseases (eg, hypertension and other cardiovascular diseases, diabetes and hypothyroidism) and prescribed medication (excluding contraception), shift work and self-disclosed use of anabolic steroids or other doping agents.	59/45 (76%)	23 M/22 F	28±5
Jurov 2022; J Int Soc Sports Nutr[76]	5	Inc: Males, 18–35 years, well trained with VO2max 55–64,9 mL/kg/min/ performance level 3 or more, BMI 19–25 kg/m^2 with body fat 5%–20%, stable body mass for the last 12 months Exc: Acute disease or chronic disease in relapse, current injuries within 3 months, undertaking specific diet	18/11 (61%)	11 M/0 F	27.5±5.6
Jurov 2022; Eur J Appl Physiol[74]	5		18/12 (67%)	12 M/0 F	27.5±5.7
Kojima 2022; PLOS ONE[65]	6	Inc: Active males. Exc: Not defined	10/10 (100%)	10 M/0 F	21.4±0.6
Kojima 2020; Med Sci Sport Exerc[66]	6	Inc: Male long-distance runners regularly involved with running-specific training (2 hours/day, 6 day/week) Exc: Not defined	8/7 (88%)	7 M/0 F	19.9±1.1
Kuikman 2025; Med Sci Sport Exerc[67]	6	Inc: Endurance-trained athletes; females were combined monophasic contraceptive pill used within 3 months before the study. Exc: Not defined	20/20 (100%)	10 M/10 F	Males: 37.8±8.9 Females: 33.1±6.9

Continued

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 2 Continued

Author and year; Journal	PEDro score	Inclusion/exclusion Criteria	N enrolled/ N Analysed (Attrition %)	Sexes (male/female)	Ages (years)
Mathisen 2020; Scand J Med Sci Sports*[68]	4	Inc: Female athletes: Ages 18–40 years, female fitness athletes, planning to compete in the upcoming season. Female references: Ages 18–40, BMI 17.5–30 kg/m^2, regularly physically active for at least two sessions/week Exc (references): planning to become pregnant, former competitive fitness/bodybuilding experience, or personal trainer or instructor in fitness industry	51/41 (80%)	0 M/41 F	28.1±5.5
Oxfeldt 2023; J Physiol[29]	7	Inc: Healthy females aged 18–30 years with a BMI between 18.5 and 30 kg/m^2, exercising 4–10 times per week, with at least one resistance and one aerobic-based or interval-based training session per week, not using any form of hormonal contraceptive ≤6 months. Exc: Irregular menstruation ≤6 months, markers of LEA (LEAF-Q≥8; EDE-Q<2.3 and RMR ratio <0.90), donated blood ≤1 month, not weight stable ≤6 months, illness/disease/injury affecting performance, smokers, vegans, current pregnancy	30/30 (100% for this review; 93% overall)	0 M/30 F	26±3
Oxfeldt 2024; Med Sci Sport Exerc[69]	7				
Schaal 2017; Scand J Med Sci Sports[70]	3	Inc: Well-trained synchronised swimmer volunteers, otherwise, not defined. Exc: Not defined	11/9 (82%)	0 M/9 F	20.4±0.4
Tektunalı Akman 2024; Front Public Health*[77]	4	Inc: Not defined. Exc: Not defined	100/83 (83%)	0 M/83 F	17.2±2.0
Zabriskie 2019; Nutrients*[78]	4	Inc: NCAA female lacrosse athletes. Exc: Any athlete who became injured and was unable to fully participate in team activities.	22/20 (91%)	0 M/20 F	20.4±1.8

Extracted data from the included studies pertaining to key study and participant details.
All standard journal abbreviations included and thus not individually defined.
*Indicates that authors were contacted for included data.
BMI, Body Mass Index; EDE-Q, Eating Disorder Examination Questionnaire; EEE, Estimated Energy Expenditure; EI, Energy Intake; Exc, Exclusion; F, Female; FFM, Fat-Free Mass; Inc, Inclusion; LEA, Low Energy Availability; LEAF-Q, Low Energy Availability in Females Questionnaire; M, Male; NCAA, National Collegiate Athletic Association; RMR, Resting Metabolic Rate; PEDro Scale, Physiotherapy Evidence Database Scale; VO2max, Maximal Aerobic Capacity.

due to differences in exact LEA criteria and thresholds, overall study design heterogeneity and/or isolated findings, pooled estimates of LEA effects on neuromusculoskeletal responses were not possible, which supports the need for more standardisation in assessment approaches and LEA measurement and threshold criteria.[80] This systematic review sets the foundation for future research examining specific neuromusculoskeletal responses to training among those with LEA, and for LEA-focused interventions to combat unfavourable training responses. The synthesised findings have key implications for clinical investigations into LEA-related barriers to recovery following major injury and surgery, such as anterior cruciate ligament reconstruction, where clinicians specifically target increased strength and muscle mass over time. This review provides a framework for established practices and methodologies for examining neuromusculoskeletal responses to training stimuli over time associated with LEA.

One of the primary desired outcomes of resistance training is to induce muscle hypertrophy, which directly correlates with muscle strength gains. In contrast, endurance-based training stimulates the performance of slow-twitch muscle fibres to perform sustained exercise and leads to increased lean muscle mass.[81] Thus, threats to lean muscle mass response to exercise are poised to hinder desired responses to training stimuli. Previous research has consistently identified reduced lean muscle mass at singular time points among those with LEA when compared with healthy expected values.[4] Yet, this review is the first to synthesise the evidence on training adaptations over time in relation to LEA status. The current emerging research suggests that those with LEA either have an impaired response and subsequently decreased FFM or lean muscle mass over time in response to training[29 60–64 66 67 69 76] or have no changes in FFM or lean muscle mass over time.[14 65 70–75 77 78] Although the evidence is conflicting, both outcomes suggest that the training stimuli are not inducing the desired adaptation of an expected increase in muscle mass due to the internal contributory factor of neuromuscular response associated with LEA. One potential underlying factor influencing the varied findings may be the duration of exposure to LEA status, as these ranged across studies from several days to several months. Length of exposure to LEA, otherwise known as prolonged LEA with an increased risk of developing REDs,[4] is poised to influence long-range changes in neuromusculoskeletal measures. Differences in LEA definition criteria and thresholds may have also influenced the variable findings across studies. The findings related to FFM and lean muscle mass should, therefore, be interpreted in this context of duration of LEA exposure and study-specific criteria and definitions outlined in table 1.

The mechanism of net protein accumulation largely occurs through muscle protein synthesis (anabolism) that exceeds muscle protein breakdown (catabolism) to rebuild from exercise-induced muscle microdamage.[82] Thus, when the body is in a prolonged energy-deficient state and key physiological and metabolic processes are slowed (or halted in extreme cases), it is highly plausible that muscle protein catabolism exceeds or is closer to anabolism, thereby impairing tissue growth.[4] Indeed, one of the included studies identified that muscle protein synthesis was significantly impaired over time among those with LEA, supporting this potential mechanistic contributor to impaired neuromusculoskeletal responses to training,[29] though this finding was in isolation and should be replicated to support this mechanistic association. Furthermore, in the absence of muscle glycogen stores, as is often evident with LEA,[63 66 69] the body will convert to alternate substrate utilisation, including the breakdown of fat and muscle tissue for the source of energy

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

during exercise. These processes may have influenced the largely negative or null influence of LEA on FFM/ lean muscle mass in the included studies for this review; however, this hypothesis is speculative.

Several key methodological differences among the included studies may have contributed to the conflicting findings of FFM/ lean muscle mass changes over time. While lean muscle mass is directly related to muscle tissue changes over time, FFM includes the mass of all tissues apart from fat and, therefore, provides an indirect estimate of muscle tissue responses to exercise. FFM was the primary outcome measure of the included studies, which may have contributed to the conflicting results. Future research should examine muscle tissue properties more specifically, and the subsequent response to training with LEA, to more directly understand the implications of underfuelling in tissue-level response. Another key consideration is the length of exposure to training and the time between assessments, as there was a wide range of assessment time frames. Although there is no direct consensus on the duration and number of sessions requisite to induce muscle hypertrophy, most changes occur only after multiple months to years,[83] and training exposure in some included studies may have been too short to induce measurable changes using DXA. Finally, the types of exercise exposures varied widely from strictly resistance training, strictly endurance training, or a combination of the two types of exposures. The current state of the evidence strongly supports the need for more standardised approaches to examining muscle responses to training. We recommend future work to conduct controlled exercise programming and more direct measures of muscle tissue response to training over time among those with and without LEA to provide more concrete evidence on training effects.

Although few studies examined more functional or sport-specific neuromusculoskeletal responses to training over time, all but one of these studies identified preliminary markers of impaired performance adaptations to training over time under LEA conditions.[62 64 69 73 74 76] Impaired sport-specific neuromusculoskeletal responses with LEA align with synthesised findings of cross-sectional studies suggesting that LEA leads to impaired endurance, aerobic capacity and strength measures.[4] These initial emerging findings of functional neuromusculoskeletal impairments warrant consideration and further study as athletes' underlying LEA may counteract the extensive strength and conditioning resources and time that sport programmes devote to athletes to enhance performance. Furthermore, there is a high likelihood that worse neuromusculoskeletal performance over time may impede the ability to continue desired sport activities and ultimately contribute to physical activity attrition. Athletes, coaches and clinicians should be aware of the consequences of underfuelling on overall health, sport performance and recovery from injury, specifically on the neuromusculoskeletal implications for stagnant or reduced responses to sport training or rehabilitation over time. Education and behaviour-based interventions, particularly those provided by credentialed clinicians with expertise in sport nutrition, consisting of both nutritional content specific to athletics as well as information about LEA and its potential impact on sport performance and health, have been demonstrated to be successful in improving nutrition knowledge and fuelling behaviours.[84] Additionally, a multidisciplinary team approach including mental health professionals to help address psychosocial factors and body dissatisfaction should be considered, given that athletes' beliefs and behaviours related to body image may mediate effects of nutrition education interventions.[5] Future work should determine if multimodal interventions including nutrition education, psychosocial and body image targeted interventions are effective in mitigating LEA and its impact on neuromusculoskeletal response to training.

One study included in this systematic review used the LEAF-Q instrument as a surrogate indicator of LEA.[68] The LEAF-Q has previously demonstrated construct validity in prior investigations assessing LEA and associated signs/symptoms in athletes.[46] When applied in the appropriate sport context, well-designed and tested surveys can effectively assess long-term energy status and overall health.[80] Such survey tools provide the potential to expand the scope of research endeavours and broaden subject recruitment in the field. However, survey-based indicators may be limited in the assessment of duration, frequency and severity of LEA exposure and must be interpreted judiciously.[4]

Future opportunities and next steps

While multiple studies fit the inclusion criteria for the present systematic review, there are many opportunities for future research to expand the understanding of neuromusculoskeletal consequences associated with LEA. Most studies assessed singular neuromusculoskeletal responses to exercise, and thus, a comprehensive view of cellular-level, tissue-level and functional-level outcomes is currently lacking, yet would facilitate a better understanding of the correlations between these measures and their implications on performance. By collectively examining these outcomes, we will better understand the mechanistic and systems-level impairments associated with LEA. Further, more direct neuromuscular assessments were missing from the current literature, namely electromyography for examining muscle activation during physical activities. Previous studies have found that poor neuromuscular activation during loaded activities such as jumping and running is attributed to loading-based injuries[39–43] (eg, BSIs) that frequently occur with LEA cases.[33] Additionally, only singular studies examined strength, muscle girth (limb circumference) and other specific measures of muscle performance. There is a need to examine these factors in relation to LEA to further appreciate threats to neuromusculoskeletal health and adaptations to training over time. Finally, there was a range of LEA thresholds and definitions, and varied length of LEA statuses across the included studies. We believe this is attributed to a lack of consensus of LEA measurement and threshold determination in the literature prior to the 2023 International Olympic Committee (IOC) Statement on problematic LEA and subsequent REDs.[4 47 80] We intentionally left the LEA definition for this systematic review broad for this reason and to capture the current state of the research related to LEA neuromusculoskeletal health. This makes interpretation of collective results more difficult as shorter timeframes may result in adaptable LEA, while long-term exposures may contribute to problematic LEA and potential development of REDs. Future studies should implement standardised LEA assessments based on the best evidence assessments of primary REDs indicators outlined in the IOC statement Clinical Assessment Tool version 2 (ie, reproductive function, bone stress injury history, bone mineral density, testosterone),[47] and other related factors such as free triiodothyronine (FT3) and measured-to-predicted resting metabolic rate ratio.[85] Future studies should also control for length of LEA to determine if there may be a dose-response relationship to neuromusculoskeletal impairments. Determining these outcomes will better prepare clinicians, coaches, strength and conditioning specialists, athletic trainers and physical therapists working with athletes with LEA to guide clinical care and interventions to optimise the use of resources and mitigate long-term health concerns stemming from underfuelling.

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Limitations

We pre-emptively decided to include studies that looked at in situ physical activity exposures in addition to more controlled, laboratory-prescribed physical activity, which limited direct comparisons or pooled analyses across included studies. Additional heterogeneity across studies through LEA definitions/thresholds, physical activity length/duration, time between neuromusculoskeletal measures and types of neuromusculoskeletal outcomes further precluded pooled analyses. The wide heterogeneity in LEA definitions employed by the included studies reflects the lack of consensus on LEA definitions in the current body of literature, in part due to differences in energy availability levels that have been associated with health and performance effects between individuals (sexes, ages) and physiologic systems. Additionally, exact energy availability calculations have been shown to be prone to problems with accuracy due to technical complexity and intensive resource burden associated with measuring energy intake, exercise energy expenditure and other contributing components (FFM, resting metabolic rate).[4 47] Future research should seek to standardise LEA assessment protocols and methodologies to reduce error and facilitate cross-study alignment under the updated clinical recommendations.[4 47] We believe that the heterogeneity of neuromusculoskeletal outcomes is reflective of the current state of the literature and highlights the need for examining more functional and neuromuscular endpoints, such as muscle electromyography during sport-related tasks, for example, in response to training by LEA status. However, we believe our overall approach to this review yielded a more robust representation of the current state of the literature. Although the majority of included studies directly measured or manipulated LEA status in laboratory settings, several studies performed LEA estimates using a field-based design (n=3) which may have presented some inaccuracies in estimation. Future studies should implement current best evidence approaches to clinical measurements for LEA.[47 80] While we attempted to contact authors to obtain information relevant to the review, we unfortunately received a low response rate (27%) which may have introduced study selection bias. The included studies had a wide range of methodological quality most commonly attributed to study design, lack of key study participant details and low retention rates; thus, some of the study findings included in this review should be interpreted cautiously.

CONCLUSIONS

This systematic review identified 21 studies that examined the effects of LEA on neuromusculoskeletal adaptations in response to physical training stimuli over time. The included studies identified that those with LEA had altered neuromusculoskeletal responses to training stimuli over time, particularly null or impaired adaptations of lean or FFM and impaired functional performance. Current evidence supports the need for future standardised and comprehensive neuromusculoskeletal response assessments combined with LEA, with particular opportunities to examine strength and muscle activation changes over time. Coaches, athletes and clinicians should be aware of the current threats to training responses associated with underfuelling to achieve desired responses to training stimuli, and ultimately improve neuromusculoskeletal health for those with LEA.

Acknowledgements We thank Talicia Tarver, MLIS, MEd, within the Virginia Commonwealth University Health Sciences Library system, for their assistance in refining and adapting the systematic review search strategy across databases.

Contributors AFDL: contributed to study question conceptualisation, registered the review, devised the search strategy, completed the search, screened studies for inclusion, methodological quality assessment, extracted data, drafted and revised the paper and served as the guarantor. KLS: contributed to study question conceptualisation, screened studies for inclusion, methodological quality assessment and revised the paper. GAD: contributed to study question conceptualisation, screened studies for inclusion, methodological quality assessment and revised the paper. KEH: contributed to study question conceptualisation, screened studies for inclusion, methodological quality assessment and revised the paper. JRO: contributed to study question conceptualisation, screened studies for inclusion, methodological quality assessment and revised the paper. LZI: contributed to study question conceptualisation and revised the paper. KEW: contributed to study question conceptualisation, devised the search strategy, served as the tie-breaker for study inclusion decision-making and revised the paper.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests AFDL receives research support from the Building Interdisciplinary Research Careers in Women's Health (BIRCWH) through the National Institute of Arthritis and Musculoskeletal and Skin Diseases of the National Institutes of Health (5K12AR084233).

Patient and public involvement statement The author group is actively involved in clinical care and community outreach partnerships involving athletes and coaches involved in sports in which low energy availability has been demonstrated to be highly prevalent in prior literature. Qualitative discussions with these athlete and coaching populations helped to inform the study's aims. In addition, the study findings are planned to be widely disseminated to these groups after publication for education purposes.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data sharing not applicable as no datasets generated and/or analysed for this study. All data presented in this systematic review were obtained from published research or obtained from direct requests to the corresponding authors of the included studies.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

ORCID iDs
Alexandra F DeJong Lempke http://orcid.org/0000-0002-5702-9184
Jessie R Oldham http://orcid.org/0000-0003-2801-2110
Kristin E Whitney http://orcid.org/0000-0002-1612-3852

REFERENCES

1 Areta JL, Taylor HL, Koehler K. Low energy availability: history, definition and evidence of its endocrine, metabolic and physiological effects in prospective studies in females and males. *Eur J Appl Physiol* 2021;121:1–21.
2 Logue DM, Madigan SM, Melin A, *et al*. Low Energy Availability in Athletes 2020: An Updated Narrative Review of Prevalence, Risk, Within-Day Energy Balance, Knowledge, and Impact on Sports Performance. *Nutrients* 2020;12:835.
3 Logue D, Madigan SM, Delahunt E, *et al*. Low Energy Availability in Athletes: A Review of Prevalence, Dietary Patterns, Physiological Health, and Sports Performance. *Sports Med* 2018;48:73–96.
4 Mountjoy M, Ackerman KE, Bailey DM, *et al*. 2023 International Olympic Committee's (IOC) consensus statement on Relative Energy Deficiency in Sport (REDs). *Br J Sports Med* 2023;57:1073–97.
5 Jagim AR, Fields J, Magee MK, *et al*. Contributing Factors to Low Energy Availability in Female Athletes: A Narrative Review of Energy Availability, Training Demands, Nutrition Barriers, Body Image, and Disordered Eating. *Nutrients* 2022;14:986.
6 Slater J, Brown R, McLay-Cooke R, *et al*. Low Energy Availability in Exercising Women: Historical Perspectives and Future Directions. *Sports Med* 2017;47:207–20.
7 Condo D, Lohman R, Kelly M, *et al*. Nutritional Intake, Sports Nutrition Knowledge and Energy Availability in Female Australian Rules Football Players. *Nutrients* 2019;11:971.
8 Ackerman KE, Holtzman B, Cooper KM, *et al*. Low energy availability surrogates correlate with health and performance consequences of Relative Energy Deficiency in Sport. *Br J Sports Med* 2019;53:628–33.

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

9 Drew M, Vlahovich N, Hughes D, *et al*. Prevalence of illness, poor mental health and sleep quality and low energy availability prior to the 2016 Summer Olympic Games. *Br J Sports Med* 2018;52:47–53.
10 Høeg TB, Olson EM, Skaggs K, *et al*. Prevalence of Female and Male Athlete Triad Risk Factors in Ultramarathon Runners. *Clin J Sport Med* 2022;32:375–81.
11 Whitney KE, DeJong Lempke AF, Stellingwerff T, *et al*. Boston Marathon athlete performance outcomes and intra-event medical encounter risk associated with low energy availability indicators. *Br J Sports Med* 2025;59:222–30.
12 Sygo J, Coates AM, Sesbreno E, *et al*. Prevalence of Indicators of Low Energy Availability in Elite Female Sprinters. *Int J Sport Nutr Exerc Metab* 2018;28:490–6.
13 Garron T, Klein DJ. Male Army ROTC Cadets Fail to Meet Military Dietary Reference Intakes and Exhibit a High Prevalence of Low Energy Availability and Poor Sleep Quality. *J Funct Morphol Kinesiol* 2023;8:95.
14 Jesus F, Castela I, Silva AM, *et al*. Risk of Low Energy Availability among Female and Male Elite Runners Competing at the 26th European Cross-Country Championships. *Nutrients* 2021;13:873.
15 Mountjoy M, Sundgot-Borgen J, Burke L, *et al*. International Olympic Committee (IOC) Consensus Statement on Relative Energy Deficiency in Sport (RED-S): 2018 Update. *Int J Sport Nutr Exerc Metab* 2018;28:316–31.
16 Mountjoy M, Sundgot-Borgen J, Burke L, *et al*. The IOC consensus statement: beyond the Female Athlete Triad—Relative Energy Deficiency in Sport (RED-S). *Br J Sports Med* 2014;48:491–7.
17 Martin D, Papageorgiou M, Colgan H, *et al*. The effects of short-term low energy availability, achieved through diet or exercise, on cognitive function in oral contraceptive users and eumenorrheic women. *Appl Physiol Nutr Metab* 2021;46:781–9.
18 Tchanturia K, Davies H, Roberts M, *et al*. Poor cognitive flexibility in eating disorders: examining the evidence using the Wisconsin Card Sorting Task. *PLoS One* 2012;7:e28331.
19 Guillaume S, Gorwood P, Jollant F, *et al*. Impaired decision-making in symptomatic anorexia and bulimia nervosa patients: a meta-analysis. *Psychol Med* 2015;45:3377–91.
20 Allen KL, Byrne SM, Hii H, *et al*. Neurocognitive functioning in adolescents with eating disorders: A population-based study. *Cogn Neuropsychiatry* 2013;18:355–75.
21 Elliott-Sale KJ, Tenforde AS, Parziale AL, *et al*. Endocrine Effects of Relative Energy Deficiency in Sport. *Int J Sport Nutr Exerc Metab* 2018;28:335–49.
22 Fahrenholtz IL, Sjödin A, Benardot D, *et al*. Within-day energy deficiency and reproductive function in female endurance athletes. *Scand J Med Sci Sports* 2018;28:1139–46.
23 Vanheest JL, Rodgers CD, Mahoney CE, *et al*. Ovarian suppression impairs sport performance in junior elite female swimmers. *Med Sci Sports Exerc* 2014;46:156–66.
24 Kaiserauer S, Snyder AC, Sleeper M, *et al*. Nutritional, physiological, and menstrual status of distance runners. *Med Sci Sports Exerc* 1989;21:120–5.
25 Rickenlund A, Thorén M, Carlström K, *et al*. Diurnal profiles of testosterone and pituitary hormones suggest different mechanisms for menstrual disturbances in endurance athletes. *J Clin Endocrinol Metab* 2004;89:702–7.
26 Williams NI, Young JC, McArthur JW, *et al*. Strenuous exercise with caloric restriction: effect on luteinizing hormone secretion. *Med Sci Sports Exerc* 1995;27:1390–8.
27 Baer JT. Endocrine parameters in amenorrheic and eumenorrheic adolescent female runners. *Int J Sports Med* 1993;14:191–5.
28 Strock NCA, De Souza MJ, Williams NI. Eating behaviours related to psychological stress are associated with functional hypothalamic amenorrhoea in exercising women. *J Sports Sci* 2020;38:2396–406.
29 Oxfeldt M, Phillips SM, Andersen OE, *et al*. Low energy availability reduces myofibrillar and sarcoplasmic muscle protein synthesis in trained females. *J Physiol* 2023;601:3481–97.
30 Areta JL, Burke LM, Camera DM, *et al*. Reduced resting skeletal muscle protein synthesis is rescued by resistance exercise and protein ingestion following short-term energy deficit. *Am J Physiol Endocrinol Metab* 2014;306:E989–97.
31 Pasiakos SM, Vislocky LM, Carbone JW, *et al*. Acute energy deprivation affects skeletal muscle protein synthesis and associated intracellular signaling proteins in physically active adults. *J Nutr* 2010;140:745–51.
32 Costill DL, Flynn MG, Kirwan JP, *et al*. Effects of repeated days of intensified training on muscle glycogen and swimming performance. *Med Sci Sports Exerc* 1988;20:249–54.
33 Heikura IA, Uusitalo ALT, Stellingwerff T, *et al*. Low Energy Availability Is Difficult to Assess but Outcomes Have Large Impact on Bone Injury Rates in Elite Distance Athletes. *Int J Sport Nutr Exerc Metab* 2018;28:403–11.
34 Ackerman KE, Cano Sokoloff N, De Nardo Maffazioli G, *et al*. Fractures in Relation to Menstrual Status and Bone Parameters in Young Athletes. *Med Sci Sports Exerc* 2015;47:1577–86.
35 Barrack MT, Gibbs JC, De Souza MJ, *et al*. Higher incidence of bone stress injuries with increasing female athlete triad-related risk factors: a prospective multisite study of exercising girls and women. *Am J Sports Med* 2014;42:949–58.
36 Hector AJ, McGlory C, Damas F, *et al*. Pronounced energy restriction with elevated protein intake results in no change in proteolysis and reductions in skeletal muscle protein synthesis that are mitigated by resistance exercise. *FASEB J* 2018;32:265–75.
37 Nichols JF, Rauh MJ, Barrack MT, *et al*. Bone mineral density in female high school athletes: interactions of menstrual function and type of mechanical loading. *Bone* 2007;41:371–7.
38 Johansson H, Kanis JA, Odén A, *et al*. A meta-analysis of the association of fracture risk and body mass index in women. *J Bone Miner Res* 2014;29:223–33.
39 Derrick TR, Dereu D, McLean SP. Impacts and kinematic adjustments during an exhaustive run. *Med Sci Sports Exerc* 2002;34:998–1002.
40 Derrick TR, Hamill J, Caldwell GE. Energy absorption of impacts during running at various stride lengths. *Med Sci Sports Exerc* 1998;30:128–35.
41 Edwards WB. Modeling Overuse Injuries in Sport as a Mechanical Fatigue Phenomenon. *Exerc Sport Sci Rev* 2018;46:224–31.
42 Edwards WB, Derrick TR, Hamill J. Musculoskeletal Attenuation of Impact Shock in Response to Knee Angle Manipulation. *J Appl Biomech* 2012;28:502–10.
43 Mizrahi J, Verbitsky O, Isakov E. Shock accelerations and attenuation in downhill and level running. *Clin Biomech (Bristol, Avon)* 2000;15:15–20.
44 Tornberg ÅB, Melin A, Koivula FM, *et al*. Reduced Neuromuscular Performance in Amenorrheic Elite Endurance Athletes. *Med Sci Sports Exerc* 2017;49:2478–85.
45 Ardern CL, Büttner F, Andrade R, *et al*. Implementing the 27 PRISMA 2020 Statement items for systematic reviews in the sport and exercise medicine, musculoskeletal rehabilitation and sports science fields: the PERSiST (implementing Prisma in Exercise, Rehabilitation, Sport medicine and SporTs science) guidance. *Br J Sports Med* 2022;56:175–95.
46 Melin A, Tornberg AB, Skouby S, *et al*. The LEAF questionnaire: a screening tool for the identification of female athletes at risk for the female athlete triad. *Br J Sports Med* 2014;48:540–5.
47 Stellingwerff T, Mountjoy M, McCluskey WT, *et al*. Review of the scientific rationale, development and validation of the International Olympic Committee Relative Energy Deficiency in Sport Clinical Assessment Tool: V.2 (IOC REDs CAT2)—by a subgroup of the IOC consensus on REDs. *Br J Sports Med* 2023;57:1109–21.
48 Maher CG, Sherrington C, Herbert RD, *et al*. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Phys Ther* 2003;83:713–21.
49 de Morton NA. The PEDro scale is a valid measure of the methodological quality of clinical trials: a demographic study. *Aust J Physiother* 2009;55:129–33.
50 Kettunen O, Mikkonen R, Linnamo V, *et al*. Nutritional intake and anthropometric characteristics are associated with endurance performance and markers of low energy availability in young female cross-country skiers. *J Int Soc Sports Nutr* 2023;20:2226639.
51 Bushmanova EA, Lyudinina AY, Bojko ER. The Prevalence of Low Energy Availability in Cross-Country Skiers during the Annual Cycle. *Nutrients* 2024;16:2279.
52 Viner RT, Harris M, Berning JR, *et al*. Energy Availability and Dietary Patterns of Adult Male and Female Competitive Cyclists With Lower Than Expected Bone Mineral Density. *Int J Sport Nutr Exerc Metab* 2015;25:594–602.
53 Stenqvist TB, Melin AK, Torstveit MK. Relative Energy Deficiency in Sport (REDs) Indicators in Male Adolescent Endurance Athletes: A 3-Year Longitudinal Study. *Nutrients* 2023;15:5086.
54 Willingham BD, Daou M, VanArsdale J, *et al*. Energy Availability in Female Collegiate Beach Volleyball Athletes. *J Strength Cond Res* 2024;38:1941–50.
55 Keay N, Francis G, Entwistle I, *et al*. Clinical evaluation of education relating to nutrition and skeletal loading in competitive male road cyclists at risk of relative energy deficiency in sports (RED-S): 6-month randomised controlled trial. *BMJ Open Sport Exerc Med* 2019;5:e000523.
56 Ihalainen JK, Kettunen O, McGawley K, *et al*. Body Composition, Energy Availability, Training, and Menstrual Status in Female Runners. *Int J Sports Physiol Perform* 2021;16:1043–8.
57 Miyamoto M, Hanatani Y, Shibuya K. Dietary intake and menstrual cycle changes in international level young athletes. *J Sports Med Phys Fitness* 2021;61:851–6.
58 Otte CJ, Mantzioris E, Salagaras BS, *et al*. Seasonal health tracking of Australian Football League Women's athletes. *Sci Med Footb* 2024;8:103–11.
59 Reed JL, De Souza MJ, Williams NI. Changes in energy availability across the season in Division I female soccer players. *J Sports Sci* 2013;31:314–24.
60 Alemany JA, Nindl BC, Kellogg MD, *et al*. Effects of dietary protein content on IGF-I, testosterone, and body composition during 8 days of severe energy deficit and arduous physical activity. *J Appl Physiol* 2008;105:58–64.
61 Friedl KE, Moore RJ, Martinez-Lopez LE, *et al*. Lower limit of body fat in healthy active men. *J Appl Physiol* 1994;77:933–40.
62 Gifford RM, Greeves JP, Wardle SL, *et al*. Measuring the Exercise Component of Energy Availability during Arduous Training in Women. *Med Sci Sports Exerc* 2021;53:860–8.
63 Ishibashi A, Kojima C, Tanabe Y, *et al*. Effect of low energy availability during three consecutive days of endurance training on iron metabolism in male long distance runners. *Physiol Rep* 2020;8:e14494.
64 Jeppesen JS, Caldwell HG, Lossius LO, *et al*. Low energy availability increases immune cell formation of reactive oxygen species and impairs exercise performance in female endurance athletes. *Redox Biol* 2024;75:103250.
65 Kojima C, Ishibashi A, Ebi K, *et al*. Exogenous glucose oxidation during endurance exercise under low energy availability. *PLoS One* 2022;17:e0276002.
66 Kojima C, Ishibashi A, Tanabe Y, *et al*. Muscle Glycogen Content during Endurance Training under Low Energy Availability. *Med Sci Sports Exerc* 2020;52:187–95.

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

67 Kuikman MA, Smith E, Mckay AKA, *et al*. Impact of Acute Dietary and Exercise Manipulation on Next-Day RMR Measurements and DXA Body Composition Estimates. *Med Sci Sports Exerc* 2025;57:285–95.
68 Mathisen TF, Rosenvinge JH, Friborg O, *et al*. Is physical exercise and dietary therapy a feasible alternative to cognitive behavior therapy in treatment of eating disorders? A randomized controlled trial of two group therapies. *Int J Eat Disord* 2020;53:574–85.
69 Oxfeldt M, Marsi D, Christensen PM, *et al*. Low Energy Availability Followed by Optimal Energy Availability Does Not Benefit Performance in Trained Females. *Med Sci Sports Exerc* 2024;56:902–16.
70 Schaal K, Tiollier E, Le Meur Y, *et al*. Elite synchronized swimmers display decreased energy availability during intensified training. *Scand J Med Sci Sports* 2017;27:925–34.
71 Baranauskas MN, Miller B, Olson JT, *et al*. Differential in Maximal Aerobic Capacity by Sex in Collegiate Endurance Athletes Consuming a Marginally Low Carbohydrate Diet. *J Am Coll Nutr* 2017;36:370–7.
72 Burke LM, Whitfield J, Ross MLR, *et al*. Short Severe Energy Restriction with Refueling Reduces Body Mass without Altering Training-Associated Performance Improvement. *Med Sci Sports Exerc* 2023;55:1487–98.
73 Barreto G, de Oliveira LF, Saito T, *et al*. Reduced Endurance Capacity and Suboptimal Energy Availability in Top-Level Female Cyclists. *Int J Sports Physiol Perform* 2021;16:1194–203.
74 Jurov I, Keay N, Spudić D, *et al*. Inducing low energy availability in trained endurance male athletes results in poorer explosive power. *Eur J Appl Physiol* 2022;122:503–13.
75 Jouhki I, Sarin HV, Jauhiainen M, *et al*. Effects of fat loss and low energy availability on the serum cardiometabolic profile of physique athletes. *Scand J Med Sci Sports* 2024;34:e14553.
76 Jurov I, Keay N, Rauter S. Reducing energy availability in male endurance athletes: a randomized trial with a three-step energy reduction. *J Int Soc Sports Nutr* 2022;19:179–95.
77 Tektunalı Akman C, Gönen Aydın C, Ersoy G. The effect of nutrition education sessions on energy availability, body composition, eating attitude and sports nutrition knowledge in young female endurance athletes. *Front Public Health* 2024;12:1289448.
78 Zabriskie HA, Currier BS, Harty PS, *et al*. Energy Status and Body Composition Across a Collegiate Women's Lacrosse Season. *Nutrients* 2019;11:470.
79 Loucks AB, Kiens B, Wright HH. Energy availability in athletes. *J Sports Sci* 2011;29 Suppl 1:S7–15.
80 Ackerman KE, Rogers MA, Heikura IA, *et al*. Methodology for studying Relative Energy Deficiency in Sport (REDs): a narrative review by a subgroup of the International Olympic Committee (IOC) consensus on REDs. *Br J Sports Med* 2023;57:1136–52.
81 Hughes DC, Ellefsen S, Baar K. Adaptations to Endurance and Strength Training. *Cold Spring Harb Perspect Med* 2018;8:a029769.
82 Lim C, Nunes EA, Currier BS, *et al*. An Evidence-Based Narrative Review of Mechanisms of Resistance Exercise–Induced Human Skeletal Muscle Hypertrophy. *Med Sci Sports Exerc* 2022;54:1546–59.
83 Damas F, Libardi CA, Ugrinowitsch C. The development of skeletal muscle hypertrophy through resistance training: the role of muscle damage and muscle protein synthesis. *Eur J Appl Physiol* 2018;118:485–500.
84 DeJong Lempke AF, Reece LM, Whitney KE. Nutrition educational interventions for athletes related to low energy availability: A systematic review. *PLoS ONE* 2025;20:e0314506.
85 De Souza MJ, Koltun KJ, Strock NC, *et al*. Rethinking the concept of an energy availability threshold and its role in the Female Athlete Triad. *Curr Opin Physiol* 2019;10:35–42.

Br J Sports Med: first published as 10.1136/bjsports-2025-110096 on 13 October 2025. Downloaded from http://bjsm.bmj.com/ on October 14, 2025 at Virginia Commonwealth University. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.